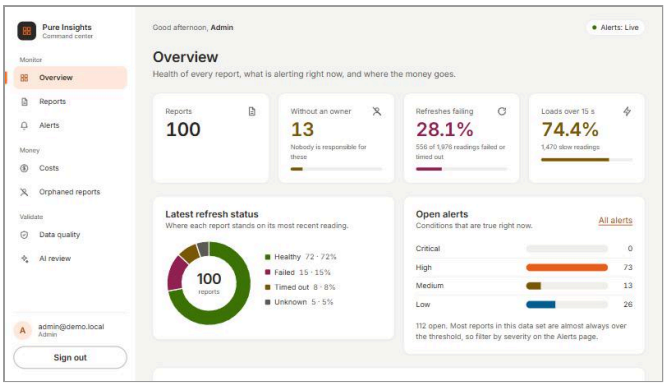
Case Study 3: Pure Insights Command Center (Platform Operational Analytics)

Submitted for	Technical Data & AI Validation Intern, Monetization (Bengaluru)
Candidate	Shivaum Shekhar Sharma, Data Science, Manipal | sshivaum@gmail.com
Domain	Full-Stack Software / Platform Engineering
Dataset	Case_Study_3_Pure_Insights_Monitoring.csv (2,000 rows, 100 reports)
Stack	React 19 (Vite), Node.js 22 / Express 5, WebSocket, JWT, Snowflake SQL, Python (independent validation)

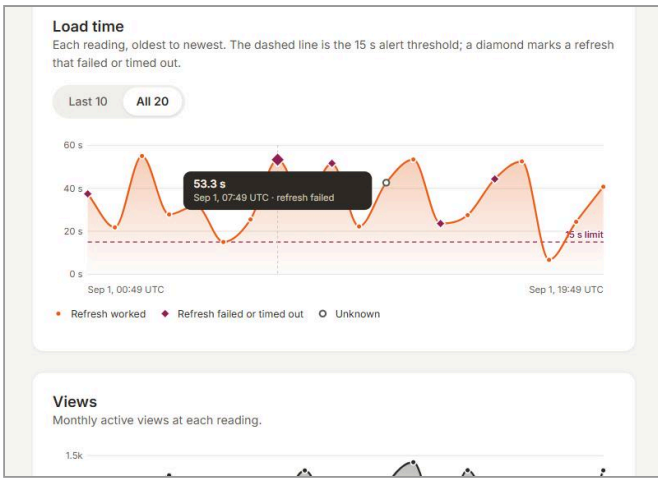
1. Summary

After the move from Tableau Server to a custom portal, the business lost visibility of report usage, refresh failures, report ownership and Snowflake compute cost. I built an admin dashboard with role-based access, alerts that arrive over a WebSocket when a report loads slowly or a refresh fails, and a suite of Snowflake SQL scripts that attribute credits to individual reports and work out the cost per active view. Every headline number was also re-derived by a separate Python program that shares no code with the application, and the two were compared item by item.

What the data and the system showed	Figure
Rows / reports / readings per report	2,000 / 100 / 20
Rows with negative Snowflake credits, set aside with a reason (not deleted)	24 (net −161.04 credits)
Reports with no owner on their latest reading	13 (242 readings have no usable owner)
Readings over the 15 s limit; of these, pinned at the 55 s ceiling	1,470 of 1,976 (74.4%); 6
Breaching conditions (slow readings plus failed refreshes; 410 readings are both, so 1,616 distinct readings) turned into alerts needing attention	2,026 into 838
Credits attributed to the 100 reports (equal to metered credits in every warehouse-hour)	1,206,352.48
Reports whose CSV credits differ from query-based attribution by more than 10%	64 of 100
Modelled spend at about $3 per credit (credits $3.62M plus AWS hosting $0.46M)	$4,074,285
Spend on the 13 orphaned reports; spend in readings with zero views	$521,801 (12.8%); $8,827
Readings whose AWS hosting cost is 3 times or more away from what their views predict	625 (31.6%)
Automated tests (all passing) / separate Python checks	499 / 68
SQL suite run on Snowflake, compared with the application's numbers	45 of 45 match (the first run found a precision bug, now fixed)



Overview page: fleet health, open alerts and a map of every report.



Report page: load time per reading with failed refreshes marked.

What I would do first

- **Assign owners to the 13 orphaned reports.** They account for $521,801 of modelled spend (12.8%, whatever the credit price) and nobody is accountable for them. The dashboard does this today, with an audit trail and undo.
- **Do not quote dollar amounts yet.** The credit price is a rough $3, and the supplied data averages about 15,000 credits per warehouse-hour while one of Snowflake's largest warehouses (6X-Large) uses 512, so absolute credits and dollars cannot be real. Rankings and ratios are usable; amounts need real usage data.
- **Agree the alert thresholds with the platform team.** 74% of readings exceed 15 s, so the rule is too noisy to be useful until the threshold and cooldown are agreed.
- **Find out whether the 24 negative credit values are adjustments or errors** before deciding how to account for them.

2. Execution tasks

Task 1: Full-stack web application with role-based access

- React front end and an Express API served from one address. Pages: sign-in, overview, reports, report detail, alerts, costs, orphaned reports, data quality.
- Sign-in returns a signed JWT (HS256, 8 hours). Three roles are enforced in the API, not only hidden in the interface:

Role	Can do
Admin	Everything: reconciliation, quarantine ledger, orphaned reports, audit trail, reassigning owners.
Owner	Health, detail, cost and alerts for the reports they currently own. Anyone else's report returns 404.
Viewer	Fleet and report health, read-only. No cost data of any kind.

- Orphaned reports: an admin sees every report with no current owner, ranked by spend, and can assign an owner. Each change is recorded in an audit trail and can be undone. A role/route matrix test and an ownership-scoping test cover every endpoint.

Task 2: Real-time telemetry and WebSocket alerting

- An alert is raised when a report's load time is over 15 s, or when an automated refresh fails or times out. Severity: over 15 s low, over 25 s medium, over 35 s high, 55 s or more critical. A failed refresh is high, three in a row is critical. A status of UNKNOWN raises nothing and resolves nothing.
- 74% of readings are over the limit, so a simple threshold would alert on almost everything. Each condition has a lifecycle: TRIGGERED (first time), ESCALATED (worse than last time, sent at once), REPEATED (still true after a 6-hour cooldown), RESOLVED (two clear readings in a row). Held-back repeats are counted on the next alert. Result: 2,026 breaching conditions (1,470 slow readings plus 556 failed refreshes; 410 readings are both) become 838 alerts that need attention. Together with 343 resolutions, that makes the 1,181 alert events.
- Alerts are pushed over a WebSocket and filtered by role on the server, so an owner never receives another owner's alerts. The server replays the cleaned readings at 10 per second to simulate telemetry.

- A client that reconnects sends the last alert number and stream id it saw and receives exactly the alerts it missed, once each, then live ones. If events were dropped, or the server restarted, it is told and refetches over REST. Without the stream id, a restart would make every new alert look already seen.
- Every one of the 1,181 alert events the engine produced was matched against a separate Python implementation of the same rules.

Task 3: Cloud cost allocation engine

- Snowflake meters credits per warehouse (billed by the second while it runs, reported hourly), not per query. Each hour's credits are therefore split across the queries that ran in it in proportion to their elapsed time, and a report's credits are the sum over its hours. Failed and timed-out queries still hold the warehouse, so they keep their share. Credits are conserved: the shares in an hour add up to that hour's metered credits.
- Cost per report = attributed credits × credit price + AWS hosting cost. Cost per active user view = that cost divided by total monthly active views (pooled over a report's readings, so one reading with very few views cannot distort it). Spend in readings with zero views is shown separately as idle cost. The data has no price for a Snowflake credit, so I used a round $3 per credit as a rough estimate. Dollar totals are good for comparing reports with each other, not as exact amounts.
- Eight reconciliation checks (R1 to R8) re-derive and cross-check the numbers from the raw tables. Results are in section 5. The full scripts are in Appendix B.
- **How the simulated usage logs were built.** The CSV has no warehouse, hour or query-time fields, so these were derived from it by fixed rules. A report's k-th reading falls in the k-th hour of 1 September 2026 (20 hours). Each report is assigned to one of four warehouses by a stable hash of its id. A warehouse-hour's metered credits are the sum of the CSV credits of the readings in it. A reading's query time is its query count times its average load time, which does not use the credits column. So the credit totals match the CSV by construction, but each report's share depends only on workload, which is why R8 can disagree with the CSV's figures. The query table has one row per report per hour, not one per query.
- The attribution is tested on small hand-worked fixtures where the right answer is known in advance (failed queries, zero views, a price change, orphaned reports, the hosting-cost cases), and each reconciliation check has a test that makes it fail on purpose.
- The question's "active user counts versus AWS hosting cost" discrepancy: the fleet's average hosting cost per view is computed, and a reading is flagged when its cost is 3 times above or below what its views predict, or when it has cost but no views. The Costs page plots every reading against that average with the 3x band.

3. Injected data anomalies and how each was handled

Anomaly	What was found	Handling
Orphaned reports (missing owner or ORPHANED_REPORT)	242 readings have no usable owner (125 empty or NaN, 117 the text ORPHANED_REPORT). 13 reports have none on their latest reading.	Flagged and kept, never dropped. The last known owner is kept as a suggestion. An admin assigns an owner with an audit trail and undo.
Negative Snowflake credits	24 rows, from −13.6 to −0.8 credits.	Set aside in a quarantine ledger with a reason code and excluded from cost totals. Nothing is deleted.
Latency spikes up to 55 s	1,470 of 1,976 readings are over 15 s, and 6 sit at exactly 55 s.	Alerted by severity tier. 55 s is critical. Repeats are de-duplicated into 838 alerts (2,026 conditions; 1,616 distinct readings).
Active users versus AWS hosting cost	625 readings are 3x or more away from the fleet's cost per view. 4 readings have zero views but real cost.	Flagged with the expected cost and the excess. Zero-view cost is reported as idle spend.

I also tested statistical anomaly detection on the raw file. All six numeric columns are spread evenly across their range, so neither the 1.5x IQR rule nor 3 standard deviations flags anything, and the 24 negative values sit well inside those limits. They are caught by a rule (credits cannot be negative) instead.

4. Submission deliverables

Deliverable 1: Runnable full-stack codebase

GitHub repository (React front end, Node.js back end, SQL scripts, tests, CI): **https://github.com/shivaumsharma/pure-insights-command-center**

To run it (Node 22 or newer): `npm install`, `npm run web:install`, `npm run data:build`, `npm run web:build`, `npm start`, then open http://localhost:3101. Demo accounts for each role are in the repository. Everything described in this document is also reproduced here, so the repository is not needed to follow it.

Deliverable 2: REST API and WebSocket technical specification

Complete, in Appendix A (authentication, roles, every endpoint, the alert object, WebSocket messages, reconnect rules, close codes). A test fails if this specification and the routes registered in the code ever disagree, and it calls each documented endpoint on a live server.

Deliverable 3: Snowflake cloud cost allocation SQL script suite

Complete, in Appendix B: the schema, then five scripts run in order (attribution, unit cost, reconciliation checks, per-report variance, views versus AWS cost).

Run on Snowflake

The first run found a bug. 42 comparisons matched and 3 did not: check R1 (attributed credits equal metered credits in every warehouse-hour) reported 52 of 80 hours unbalanced where 0 was expected, its status was FAIL, and the sum of per-report differences was 0.23 against 0.02. The total of attributed credits was also 0.19 higher out of 1.2 million, which was inside the comparison tolerance. The cause was how the workload share was calculated: dividing exact decimals keeps only a limited number of decimal places on Snowflake, while the local SQLite engine divides in floating point, so none of my local tests could show it. I changed the division to DOUBLE, added tests that fail without the change, and ran the suite again.

I ran the suite on a Snowflake trial account (version 10.37.100, region AWS_AP_SOUTHEAST_7 (AWS Asia Pacific, Thailand), as reported by Snowflake) on 14 October 2026. The schema script created the database and tables, the five generated data files were loaded, the five scripts were run in order, and each result was compared with what the application computed locally (row counts, column totals, statuses). All 45 comparisons matched: every row count, total and status that Snowflake produced equals the value the application computed locally.

Compared	Snowflake	Application	Result
11 unit cost: top report	RPT_299	RPT_299	MATCH
11 unit cost: top report total cost	71948.79	71948.79	MATCH
12 R1: actual value	0	0	MATCH
12 R1: status	PASS	PASS	MATCH
12 R2: actual value	0	0	MATCH
12 R2: status	PASS	PASS	MATCH
12 R3: actual value	0	0	MATCH
12 R3: status	PASS	PASS	MATCH
12 R4: actual value	0	0	MATCH
12 R4: status	PASS	PASS	MATCH
12 R5: actual value	0	0	MATCH
12 R5: status	PASS	PASS	MATCH
12 R6: actual value	0	0	MATCH
12 R6: status	PASS	PASS	MATCH
12 R7: actual value	1206352.48	1206352.48	MATCH
12 R7: status	INFO	INFO	MATCH

12 R8: actual value	64	64	MATCH
12 R8: status	FAIL	FAIL	MATCH
13 variance: reports outside tolerance	64	64	MATCH
14 AWS mismatch: flagged readings	625	625	MATCH
attributed credits in total	1206352.48	1206352.48	MATCH
attribution rows	1976	1976	MATCH
rows loaded: CORE.OBSERVATIONS_CLEAN	1976	1976	MATCH
rows loaded: CORE.QUARANTINE_LEDGER	24	24	MATCH
rows loaded: CORE.REPORTS	100	100	MATCH
rows loaded: SIM_ACCOUNT_USAGE.QUERY_HISTORY	1976	1976	MATCH
rows loaded: SIM_ACCOUNT_USAGE.WAREHOUSE_METERING_HISTORY	80	80	MATCH

The remaining 18 comparisons are column totals for the unit-cost, variance and hosting-cost outputs. The run used the simulated usage tables built from the CSV, not real ACCOUNT_USAGE views.

5. Evaluator rubric headings

Full-stack web architecture

- Four tiers: the React dashboard; the Express backend (REST API with JWT and three roles, alert engine, WebSocket hub); the SQL suite; and the data layer (cleaning pipeline, simulated Snowflake usage logs, exceptions ledger). The raw CSV flows through the cleaning step, which quarantines and never deletes, into the usage logs the SQL reads.
- One process serves the API, the WebSocket and the built interface, so there is no cross-origin setup. The interface is served with a strict Content-Security-Policy; other origins are refused at the WebSocket handshake.
- The cost scripts use portable SQL, so the same files run on Snowflake and on a local SQLite stand-in used for fast tests.
- Every threshold that is my choice rather than the brief's (10% credit tolerance, 3x hosting band, severity tiers, 6-hour cooldown, $3 credit price) is written down with its reason in a decision log of 46 entries.

Real-time alerting and telemetry

- Alert rules, severity, lifecycle and de-duplication are described in Task 2. The WebSocket protocol, close codes and reconnect rules are in Appendix A.
- Tested over real sockets (21 tests): authentication in the first message, role filtering, replay after a reconnect, restart detection, limits on connections and message size, and rejection of foreign origins.

Cloud cost SQL attribution

Reconciliation checks on the real data:

Check	Question	Result
R1	Do attributed credits equal metered credits in every warehouse-hour?	PASS
R2	Any metered credits in hours when no queries ran?	PASS (0)
R3 to R5	Did any join silently drop an observation or a query row?	PASS (0 each)
R6	Any negative credits left after cleaning?	PASS (0)
R7	Attributed total against the CSV total	INFO: equal, but the logs were built from the CSV, so this proves nothing by itself
R8	Per-report attributed credits against the CSV's claimed credits, within 10%	FAIL: 64 of 100 reports outside tolerance

R2 to R6 are structural guards against SQL mistakes such as a join silently dropping rows; on data built from the same CSV they can only fail if the SQL is wrong. R1 is the check that caught the Snowflake precision bug. R7 is true by construction. R8

is the one that compares two independent figures.

R8 fails because the supplied data was generated so that each report's credits are independent of its query workload. It shows that the reconciliation can catch a disagreement; it is not evidence that any real report is mis-billed. The largest gap is RPT_299: the CSV claims 7,795.11 credits and the query logs imply 22,391.93.

Code quality and system docs

- 499 automated tests (242 back end, 257 front end). The tests were mutation-checked: I broke the code deliberately in throwaway copies and confirmed a test failed each time. One case, removing a divide-by-zero guard, cannot be detected on the local engine and is recorded as such.
- GitHub Actions runs the data drift check, all tests, the build, the Python validation and two end-to-end runs that start the whole product, on a clean machine.
- The separate Python validation (68 checks) re-derives every number above from the raw CSV. It also parses every SQL script as Snowflake SQL and resolves each table and column against the schema.
- Decision log (46 entries), API specification with a drift test, and a findings log with evidence, impact, who to route each item to and the next step.

6. Findings, routed

#	Finding	Impact	Route to	Next step
1	24 rows have negative credits.	Counting them understates usage; setting them aside nets the total up by 161.04 credits.	Snowflake usage / FinOps	Confirm whether these are credit adjustments or a feed defect.
2	242 readings have no usable owner; 13 reports have none now. Their 13 reports account for $521,801 of $4,074,285 modelled spend (12.8%).	Nobody is accountable for them.	Platform admin	Assign owners; stop the feed writing the placeholder.
3	All 100 reports show more than one owner across their 20 readings.	"Current owner" is ambiguous. I define it as the owner on the latest reading.	Platform admin	Choose a source of truth for ownership.
4	74.4% of readings exceed 15 s; 6 are pinned at 55 s.	A fixed 15 s rule fires on almost everything. 55 s looks like a timeout.	Platform engineering	Confirm the threshold and what 55 s means.
5	556 refreshes failed or timed out (28.1%); 98 readings have status UNKNOWN.	UNKNOWN is neither failure nor recovery.	BI operations	Find why the status is unknown.
6	91 readings have more peak concurrent users than monthly views.	The two measures cannot both be right, or they cover different windows.	Analytics owner	Document each column's window.
7	625 readings have hosting cost 3x or more away from what their views predict; 4 have cost but no views.	A prompt to investigate, not a verdict; hosting also scales with concurrency and data volume. It flags 31.6% because this data is uncorrelated; on real data the baseline would need calibrating.	Cloud platform / FinOps	Agree a tolerance that fits how hosting is billed.
8	64 of 100 reports claim credits more than 10% from what query workload implies.	In this data the two are unrelated by construction.	FinOps	Rerun on real ACCOUNT_USAGE data, where this check matters.

9	A first run on Snowflake failed check R1 in 52 of 80 warehouse-hours.	Exact-decimal division loses precision on Snowflake; credits were off by about 0.19 in 1.2 million. Not visible on the local engine.	Whoever maintains the SQL	Fixed (division in DOUBLE), guarded by tests, rerun matched. Rerun after any change to the share formula.
10	The CSV has no timestamps and its columns are uncorrelated.	Alert order rests on a synthetic timeline; rankings reflect how the data was generated.	Data engineering	Provide a reading time per row; do not draw business conclusions from rankings.
11	The credit scale cannot be real: the metering tables average 15,079 credits per warehouse-hour (highest 18,724).	One 6X-Large Snowflake warehouse uses 512 credits an hour. Absolute credits, dollars and the top-cost report (RPT_299) are artefacts of the synthetic data; rankings and ratios still work.	Dataset owner	Check magnitudes against real usage before quoting any amount.

7. Additional work, limits and next steps

- **Explanations of alerts, checked against the data.** An alert can be explained by a language model or by plain rules. The model is not trusted: it receives a numbered evidence pack and must cite which datapoint supports each claim and the value it read. A checker compares every citation with the real data and rejects numbers that are not in the evidence. Answers that pass are shown with a list of how they were checked; answers that cannot be verified wait for an admin; answers that contradict the data are withheld, and others see the rule-based explanation instead. On 10 alerts, four free local models passed the checks 6, 3, 0 and 0 times, and the rule-based version 9 times. The rules never give a wrong answer: when the data supports no cause they hand the alert to a person (117 of the 838 alerts), which is why they do not score 10 of 10. That is a small sample.
- **Data limits.** The usage logs are simulated from the CSV, so warehouse totals equal the CSV by construction. The Snowflake run above used these simulated tables, so the reconciliation still matters on real ACCOUNT_USAGE views. Dollar amounts rest on a rough credit price (see Task 3).
- **Assumptions.** Each reading is one billing period with its own credits, hosting cost and monthly views; views are not de-duplicated across readings. Quarantining a row for negative credits also removes that reading from alerting (24 of 2,000 readings); a production version would exclude only the credit field. Alerts use an invented timeline, because the CSV has no timestamps.
- **Scope of the checks.** The Python validator follows the same written rules as the application, in a different language, so it catches coding slips and drift rather than a different reading of the brief.
- **Next.** On real data, attribute with Snowflake's QUERY_ATTRIBUTION_HISTORY view (it leaves out warehouse idle time, which would need its own line), or split each query's runtime across the hours it spans. Add a flag-only statistical screen beside the rules, a magnitude check on credits, and stored alert history so it survives a restart.

Appendix A. REST API and WebSocket specification

Base URL `http://localhost:3101`. All API routes are under `/api`, return JSON and use one error shape: `{ "error": { "code": "FORBIDDEN", "message": "...", "details": {} } }`.

Status	code	Meaning
400	BAD_REQUEST	Invalid query parameter, body or id
401	UNAUTHENTICATED / INVALID_CREDENTIALS	No, bad, expired or forged token; wrong email or password
403	FORBIDDEN	Signed in, but the role may not do this
404	NOT_FOUND	Unknown route or report, or a report outside an owner's scope
413	PAYLOAD_TOO_LARGE	Body over 10 KB
429	TOO_MANY_ATTEMPTS	Five failed sign-ins for one email and address in 15 minutes (see Retry-After)
500	INTERNAL	Generic message only; detail is logged on the server

A1. Authentication and roles

`POST /api/auth/login` with `{ "email", "password" }` returns `{ token, token_type: "Bearer", expires_in, user }`. Send `Authorization: Bearer <token>` on every other request. Tokens are HS256 JWTs (8 hours) carrying only email and role; the algorithm is pinned on verification. A wrong password and an unknown email return the same 401. `GET /api/auth/me` returns the signed-in user. `GET /healthz` is an unauthenticated liveness check.

A2. REST endpoints

List endpoints take `limit` (1 to 200, default 50) and `offset` (default 0) and return `{ data, page: { limit, offset, total } }`. Responses with dollar values add `meta: { credit_price_usd, price_is_placeholder: true }`.

Method and path	Roles	Notes
GET /api/health	all	Fleet counts: reports by latest refresh status, orphaned, refresh success and failure rates, latency breaches over 15 s. For an owner it covers only their reports.
GET /api/reports	all	Filters `status`, `orphaned`, `q`; `sort` = report_id, latency, failures, breaches, cost (not for viewers); `order`. Admin and owner rows include cost.
GET /api/reports/:id	all	Summary, up to 20 recent readings (newest first), and cost for admin and owner. An owner asking for another's report gets 404.
GET /api/costs/reports	admin, owner	Cost per report and per view, idle spend. `sort` = total_cost, cost_per_view, idle_cost; `orphaned`.
GET /api/costs/reconciliation	admin	Checks R1 to R8 with a pass / fail / info summary.
GET /api/costs/variance	admin	Attributed against CSV-claimed credits per report; `status` filter.
GET /api/costs/aws-mismatch	admin	Readings whose AWS cost does not fit their views; `type` filter.
GET /api/costs/aws-usage	admin	Every reading's views and hosting cost with its flag, unpaged, plus the fleet cost per view and the 3x ratio.
GET /api/orphans	admin	Reports with no current owner, costliest first, with total orphaned spend.
PUT /api/reports/:id/owner	admin	Body `{ "email" }`. Assigns an owner; recorded in the audit trail.
DELETE /api/reports/:id/owner	admin	Removes the assignment.
GET /api/quarantine	admin	Rows removed by cleaning, with reason codes; `reason` filter.
GET /api/dq-report, GET /api/audit	admin	The data-quality report; owner assignments, newest first.
GET /api/alerts	all	History, newest first. Filters `alert_type` (LATENCY, REFRESH_FAILURE), `severity`, `lifecycle`, `report_id`. An owner sees alerts only for reports they own at the time of the request.

GET /api/alerts/active	all	Conditions open now, worst first, with counts by severity.
GET /api/alerts/summary	admin	Readings processed, breaches, alerts per lifecycle stage, readings held back.
GET /api/alerts/:eventId/triage	all	Checked explanation of an open alert (generated on first request, then cached; 429 if asked too often).
GET /api/triage/review, POST /api/triage/:eventId/review, GET /api/triage/stats	admin	Held explanations; approve or dismiss one; how the explainers are doing.

A3. Alert object

```
{ "event_id": 42, "alert_type": "LATENCY", "lifecycle": "ESCALATED", "severity": "high",
  "report_id": "RPT_201", "report_name": "Analytics Dashboard 2", "observation_id": "RPT_201:004",
  "observed_at": "2026-09-01T03:00:30.000Z", "suppressed_since_last": 2,
  "value": 41.2, "threshold": 15, "message": "RPT_201 (Analytics Dashboard 2) load time 41.2 s is over 15
s (high)" }
```

Lifecycle: TRIGGERED, ESCALATED, REPEATED (after the 6-hour cooldown), RESOLVED (two clear readings). Refresh-failure alerts carry the status and `consecutive_failures`; RESOLVED alerts carry `active_for_ms`. Times are event time, not wall-clock time. Severity: over 15 s low, over 25 s medium, over 35 s high, 55 s or more critical; a failed refresh is high, three in a row critical.

A4. WebSocket `ws://localhost:3101/ws`

A browser WebSocket cannot set headers and a token in the URL leaks into logs, so the token is sent in the first message.

```
client -> { "type": "auth", "token": "<jwt>", "last_event_id": 41, "stream_id": "9f3a1c2e" }   // last
two optional
server -> { "type": "ready", "user": {...}, "last_event_id": 57, "stream_id": "9f3a1c2e", "resumed": true
}
server -> { "type": "gap", "reason": "dropped" | "restarted", "oldest_event_id": 50, "last_event_id": 57
}   // only if needed
server -> { "type": "alert", "event_id": 42, "alert_type": "LATENCY", ... }       // missed alerts oldest
first, then live
client -> { "type": "ping" }      server -> { "type": "pong", "server_time": "..." }
server -> { "type": "error", "code": "...", "message": "..." }                   // then the socket
closes
```

- **Reconnecting.** Remember the highest event_id processed and the stream_id from `ready`. On reconnect send both: you receive exactly the alerts you missed (only those your role may see), in order, each once, then live alerts. If `gap` arrives, events were dropped from the bounded history (2,000 alerts): refetch GET /api/alerts and GET /api/alerts/active. Reconnect with exponential backoff and jitter. Omit last_event_id on a first connection to get live alerts only.
- **Server restarts.** Event ids start again from 1 after a restart, so an id only means something within one run, which is what stream_id names. If the stream_id sent is not current (or last_event_id is beyond the newest event) the server sends `gap` with reason "restarted" and replays nothing; the client resets to the id in `ready` and refetches over REST.
- **Close codes.** 4401 not authenticated, bad or expired token, 5 s authentication timeout, or token expired while connected. 4429 more than 5 connections for one user. 1008 invalid message or too many bad messages. 1003 binary frame. 1009 message over 4 KB. 1001 server shutting down. The server pings every 30 s and drops clients that do not answer.
- **Handshake.** Only `/ws`. A browser Origin is accepted when it matches the server's own origin or is listed in ALLOWED_ORIGINS; anything else gets 403, including look-alike hosts, other ports and a null origin. Clients that send no Origin (scripts, tests) are allowed because the token authenticates them.
- **Telemetry source.** The server replays the cleaned readings at REPLAY_EVENTS_PER_SECOND (default 10) through the alert engine, then loops with timestamps shifted forward.

Appendix B. Snowflake cost allocation SQL suite

Run in order after `USE DATABASE PURE_INSIGHTS;`. Parameters (credit price 3.0 as a rough estimate, variance tolerance 10, hosting ratio 3) live in a one-row table, CORE.COST_PARAMS. The schema script (not listed in full) creates the database PURE_INSIGHTS, schemas CORE and SIM_ACCOUNT_USAGE, and six tables: CORE.OBSERVATIONS_CLEAN (1,976 rows, one per usable reading), CORE.QUARANTINE_LEDGER (24), CORE.REPORTS (100), CORE.COST_PARAMS, SIM_ACCOUNT_USAGE.QUERY_HISTORY and SIM_ACCOUNT_USAGE.WAREHOUSE_METERING_HISTORY (hourly per warehouse). The last two imitate the real ACCOUNT_USAGE views, which are read-only.

B1. 10_credit_attribution.sql: split each warehouse-hour's credits across its queries

```
-- Step 1 of 5: attribute warehouse credits to reports.
--
-- Method (the standard way to cost queries on a shared warehouse, since Snowflake meters credits per warehouse,
-- reported hourly, not per query): split each warehouse-hour's credits across the queries that ran in it, in proportion to
-- their share of elapsed time. A report's credits are the sum over the hours it ran.
--
-- Grain of the output: one row per QUERY_HISTORY row (here, one report in one hour).
--   attributed_credits   this report's share of the warehouse-hour credits
--   reported_credits     what the source CSV claims the report consumed (used only for reconciliation)
--
-- Rules:
--   * Failed and timed-out queries still hold the warehouse, so they keep their share.
--   * Credits are conserved: attributed credits in a warehouse-hour sum to that hour's metered credits.
--     A warehouse-hour with credits but no queries attributes nothing (checked in 12_reconciliation_checks.sql).
--   * LEFT JOIN to observations so a query we cannot match to an observation still carries its credits.
--
-- Portable SQL: runs on Snowflake after "USE DATABASE PURE_INSIGHTS;" and on the local SQLite warehouse.
-- Run this script first; scripts 11-14 read the table it builds.

DROP TABLE IF EXISTS core.cost_allocation;

CREATE TABLE core.cost_allocation AS
WITH workload AS (
  SELECT
    q.query_id,
    REPLACE(q.query_tag, 'report:', '') AS report_id,
    q.warehouse_name,
    q.start_time                        AS query_start,
    q.query_count,
    q.total_elapsed_time_ms             AS elapsed_ms,
    m.start_time                        AS window_start,
    m.end_time                          AS window_end,
    m.credits_used                      AS window_credits
  FROM sim_account_usage.query_history q
  JOIN sim_account_usage.warehouse_metering_history m
    ON  m.warehouse_name = q.warehouse_name
    AND q.start_time >= m.start_time
    AND q.start_time <  m.end_time
),
shares AS (
  SELECT
    w.*,
    SUM(w.elapsed_ms) OVER (PARTITION BY w.warehouse_name, w.window_start) AS window_elapsed_ms
  FROM workload w
),
attributed AS (
  SELECT
    s.*,
    -- DOUBLE on purpose: Snowflake keeps only a limited number of decimal places when it divides exact numbers, which left
shares slightly
    -- off and made check R1 fail in 52 of 80 warehouse-hours on a first run on Snowflake. Floating point balances to well under
1e-6.
    CASE WHEN s.window_elapsed_ms > 0 THEN CAST(s.elapsed_ms AS DOUBLE) / s.window_elapsed_ms ELSE 0 END AS workload_share
  FROM shares s
)
SELECT
  o.observation_id,
  a.query_id,
  a.report_id,
  o.report_name,
  a.warehouse_name,
  a.window_start,
  a.window_end,
  a.query_count,
  a.elapsed_ms,
  a.window_elapsed_ms,
  a.window_credits,
  a.workload_share,
  a.workload_share * a.window_credits AS attributed_credits,
```

```
  o.snowflake_credits_consumed        AS reported_credits,
  o.monthly_active_views,
  o.aws_hosting_cost_usd,
  o.owner_email
FROM attributed a
LEFT JOIN core.observations_clean o
  ON  o.report_id   = a.report_id
  AND o.observed_at = a.query_start;
```

B2. 11_unit_cost.sql: cost per report and per active view, idle spend

```
-- Step 2 of 5: cost per report and cost per active view.
--
--   total_cost_usd   = attributed_credits x credit_price_usd + AWS hosting cost
--   cost_per_view    = total_cost_usd / total active views   (NULL when a report has no views at all)
--
-- Definitions worth knowing:
--   * Pooled ratio: SUM(cost) / SUM(views) across a report's observations, not the average of per-observation
--     ratios. It cannot be wrecked by one observation with a handful of views.
--   * Each observation is treated as one billing period with its own cost and its own views. The source has
--     no overlap information, so views are not de-duplicated across observations (assumption D11).
--   * idle_cost_usd is spend during observations that had zero views. It is real spend with nobody using
--     the report, so it is shown separately and not hidden inside the ratio.
--   * Credits are the ATTRIBUTED credits from step 1, not the CSV's claimed figure.
--   * credit_price_usd is a placeholder in core.cost_params; do not quote dollar totals until it is replaced.
--
-- Orphan columns come from core.reports so the dashboard can show "money spent on reports nobody owns".

SELECT
  c.report_id,
  MAX(c.report_name)                                                              AS report_name,
  r.is_orphaned,
  r.last_known_owner_email,
  COUNT(*)                                                                        AS observations,
  SUM(c.monthly_active_views)                                                     AS total_views,
  ROUND(SUM(c.attributed_credits), 4)                                             AS attributed_credits,
  ROUND(SUM(c.attributed_credits) * p.credit_price_usd, 2)                        AS credit_cost_usd,
  ROUND(SUM(c.aws_hosting_cost_usd), 2)                                           AS aws_cost_usd,
  ROUND(SUM(c.attributed_credits) * p.credit_price_usd + SUM(c.aws_hosting_cost_usd), 2) AS total_cost_usd,
  CASE WHEN SUM(c.monthly_active_views) > 0
       THEN ROUND((SUM(c.attributed_credits) * p.credit_price_usd + SUM(c.aws_hosting_cost_usd))
                  / SUM(c.monthly_active_views), 4)
  END                                                                             AS cost_per_view_usd,
  CASE WHEN SUM(c.monthly_active_views) > 0
       THEN ROUND(SUM(c.attributed_credits) * p.credit_price_usd / SUM(c.monthly_active_views), 4)
  END                                                                             AS credit_cost_per_view_usd,
  SUM(CASE WHEN c.monthly_active_views = 0 THEN 1 ELSE 0 END)                     AS zero_view_observations,
  ROUND(SUM(CASE WHEN c.monthly_active_views = 0
                 THEN c.attributed_credits * p.credit_price_usd + c.aws_hosting_cost_usd
                 ELSE 0 END), 2)                                                  AS idle_cost_usd
FROM core.cost_allocation c
CROSS JOIN core.cost_params p
LEFT JOIN core.reports r ON r.report_id = c.report_id
WHERE c.observation_id IS NOT NULL
GROUP BY c.report_id, r.is_orphaned, r.last_known_owner_email, p.credit_price_usd
ORDER BY total_cost_usd DESC, c.report_id;
```

B3. 12_reconciliation_checks.sql: checks R1 to R8

```
-- Step 3 of 5: independent checks on the cost numbers. One row per check: PASS, FAIL or INFO.
--
-- Why this exists: the dashboard shows attributed credits. These checks re-derive and cross-check them
-- from the raw tables so a wrong number is caught here instead of in front of a stakeholder.
--
--   R1  Conservation per warehouse-hour: attributed credits equal metered credits (windows that had queries).
--   R2  Unattributed credits: metered credits in windows with no queries at all. Should be zero.
--   R3  Every clean observation received a cost allocation (no silent drops from the join).
--   R4  Every allocation matched an observation (no orphaned query rows carrying credits).
--   R5  Every query row landed in a metering window (nothing fell outside the hourly buckets).
--   R6  No negative credits survived cleaning.
--   R7  Attributed total vs the CSV's claimed total. INFO only: the warehouse totals were BUILT from the CSV,
--       so agreement here is true by construction and proves nothing.
--   R8  Per-report agreement: attributed vs CSV-claimed credits within tolerance. This is the real test, because
--       query workload is independent of the CSV credits. Expect FAIL on the synthetic data (see detail in 13).

WITH window_diff AS (
  SELECT
    m.warehouse_name,
    m.start_time,
    m.credits_used,
    COALESCE(SUM(c.attributed_credits), 0) AS attributed,
    COUNT(c.query_id)                      AS queries
  FROM sim_account_usage.warehouse_metering_history m
  LEFT JOIN core.cost_allocation c
```

```
    ON c.warehouse_name = m.warehouse_name AND c.window_start = m.start_time
  GROUP BY m.warehouse_name, m.start_time, m.credits_used
),
report_var AS (
  SELECT
    report_id,
    SUM(attributed_credits) AS attributed,
    SUM(reported_credits)   AS reported
  FROM core.cost_allocation
  WHERE observation_id IS NOT NULL
  GROUP BY report_id
),
checks AS (
  SELECT 'R1' AS check_id, 'Attributed = metered credits in every warehouse-hour with queries' AS description,
         0 AS expected_value,
         (SELECT COUNT(*) FROM window_diff WHERE queries > 0 AND ABS(credits_used - attributed) > 0.000001) AS actual_value,
         0 AS is_info,
         'count of warehouse-hours that do not balance' AS note
  UNION ALL
  SELECT 'R2', 'Metered credits in warehouse-hours with no queries', 0,
         (SELECT COALESCE(SUM(credits_used), 0) FROM window_diff WHERE queries = 0), 0,
         'credits that could not be attributed to any report'
  UNION ALL
  SELECT 'R3', 'Clean observations without a cost allocation', 0,
         (SELECT COUNT(*) FROM core.observations_clean o
           WHERE o.observation_id NOT IN (SELECT observation_id FROM core.cost_allocation WHERE observation_id IS NOT NULL)), 0,
         'a non-zero value means the observation-to-query join dropped rows'
  UNION ALL
  SELECT 'R4', 'Allocations that matched no observation', 0,
         (SELECT COUNT(*) FROM core.cost_allocation WHERE observation_id IS NULL), 0,
         'these carry credits but are excluded from unit cost'
  UNION ALL
  SELECT 'R5', 'Query rows outside every metering window', 0,
         (SELECT COUNT(*) FROM sim_account_usage.query_history
           WHERE query_id NOT IN (SELECT query_id FROM core.cost_allocation)), 0,
         'their elapsed time earned no credits'
  UNION ALL
  SELECT 'R6', 'Negative credits in the clean store', 0,
         (SELECT COUNT(*) FROM core.observations_clean WHERE snowflake_credits_consumed < 0), 0,
         'negatives should have been quarantined in Phase 1'
  UNION ALL
  SELECT 'R7', 'Attributed total vs CSV-claimed total (true by construction)',
         (SELECT COALESCE(SUM(snowflake_credits_consumed), 0) FROM core.observations_clean),
         (SELECT COALESCE(SUM(attributed_credits), 0) FROM core.cost_allocation), 1,
         'warehouse credits were built from the CSV, so this match is circular'
  UNION ALL
  SELECT 'R8', 'Reports whose attributed credits differ from CSV-claimed by more than the tolerance', 0,
         (SELECT COUNT(*) FROM report_var v CROSS JOIN core.cost_params p
           WHERE CASE WHEN v.reported > 0
                      THEN ABS(v.attributed - v.reported) * 100.0 / v.reported
                      ELSE CASE WHEN v.attributed > 0 THEN 1000000 ELSE 0 END
                 END > p.variance_tolerance_pct), 0,
         'the real reconciliation: query workload is independent of the CSV credits column'
)
SELECT
  check_id,
  description,
  expected_value,
  actual_value,
  actual_value - expected_value AS difference,
  CASE
    WHEN is_info = 1 THEN 'INFO'
    WHEN ABS(actual_value - expected_value) <= 0.000001 THEN 'PASS'
    ELSE 'FAIL'
  END AS status,
  note
FROM checks
ORDER BY check_id;
```

B4. 13_report_variance.sql: attributed against claimed credits per report

```
-- Step 4 of 5: per-report detail behind check R8. Attributed credits vs the credits the CSV claims.
--
-- variance_pct = (attributed - reported) / reported x 100, so a positive number means the dashboard's
-- source figure UNDER-states what the query logs imply, and a negative number means it over-states.
-- Reported = 0 with attributed > 0 has no meaningful percentage, so variance_pct is NULL and the status
-- is OUTSIDE_TOLERANCE.
--
-- Worst offenders first.

SELECT
  v.report_id,
  ROUND(v.reported, 2)   AS reported_credits,
  ROUND(v.attributed, 2) AS attributed_credits,
  ROUND(v.attributed - v.reported, 2) AS difference,
```

```
  CASE WHEN v.reported > 0 THEN ROUND((v.attributed - v.reported) * 100.0 / v.reported, 2) END AS variance_pct,
  CASE
    WHEN v.reported > 0 AND ABS(v.attributed - v.reported) * 100.0 / v.reported <= p.variance_tolerance_pct THEN
'WITHIN_TOLERANCE'
    WHEN v.reported = 0 AND v.attributed = 0 THEN 'WITHIN_TOLERANCE'
    ELSE 'OUTSIDE_TOLERANCE'
  END AS status
FROM (
  SELECT report_id, SUM(attributed_credits) AS attributed, SUM(reported_credits) AS reported
  FROM core.cost_allocation
  WHERE observation_id IS NOT NULL
  GROUP BY report_id
) v
CROSS JOIN core.cost_params p
ORDER BY ABS(v.attributed - v.reported) DESC, v.report_id;
```

B5. 14_views_vs_aws_cost.sql: hosting cost that does not fit the views

```
-- Step 5 of 5: AWS hosting cost that does not fit the number of active views.
--
-- The brief calls out "discrepancies between active user counts and AWS hosting compute costs".
-- Definition used here (active users are approximated by monthly_active_views):
--
--   fleet_aws_per_view  = total AWS cost / total views, over observations that have views
--   expected_aws_usd    = fleet_aws_per_view x this observation's views
--   cost_ratio          = actual AWS cost / expected AWS cost
--
--   COST_HIGH_FOR_USAGE      ratio >= aws_mismatch_ratio       (paying much more than usage explains)
--   COST_LOW_FOR_USAGE       ratio <= 1 / aws_mismatch_ratio   (heavy usage on a suspiciously cheap host)
--   ZERO_VIEWS_WITH_COST     no views at all but AWS cost > 0  (ratio undefined)
--
-- Only flagged observations are returned, largest overspend first. A flag is a prompt to investigate, not a
-- verdict: AWS compute scales with concurrency and data volume as well as views, and a single observation is noisy.

WITH fleet AS (
  SELECT CAST(SUM(aws_hosting_cost_usd) AS DOUBLE) / SUM(monthly_active_views) AS aws_per_view
  FROM core.observations_clean
  WHERE monthly_active_views > 0
),
scored AS (
  SELECT
    o.observation_id,
    o.report_id,
    o.observed_at,
    o.monthly_active_views,
    o.concurrent_user_peak,
    o.aws_hosting_cost_usd,
    f.aws_per_view                                  AS fleet_aws_per_view,
    f.aws_per_view * o.monthly_active_views         AS expected_aws_usd,
    CASE WHEN o.monthly_active_views > 0
         THEN o.aws_hosting_cost_usd / (f.aws_per_view * o.monthly_active_views)
    END                                             AS cost_ratio
  FROM core.observations_clean o
  CROSS JOIN fleet f
),
classified AS (
  SELECT
    s.*,
    CASE
      WHEN s.monthly_active_views = 0 AND s.aws_hosting_cost_usd > 0 THEN 'ZERO_VIEWS_WITH_COST'
      WHEN s.cost_ratio >= p.aws_mismatch_ratio                      THEN 'COST_HIGH_FOR_USAGE'
      WHEN s.cost_ratio <= 1.0 / p.aws_mismatch_ratio                THEN 'COST_LOW_FOR_USAGE'
    END AS mismatch_type
  FROM scored s
  CROSS JOIN core.cost_params p
)
SELECT
  observation_id,
  report_id,
  observed_at,
  monthly_active_views,
  concurrent_user_peak,
  ROUND(aws_hosting_cost_usd, 2)                    AS aws_cost_usd,
  ROUND(expected_aws_usd, 2)                        AS expected_aws_usd,
  ROUND(aws_hosting_cost_usd - expected_aws_usd, 2) AS excess_aws_usd,
  ROUND(cost_ratio, 2)                              AS cost_ratio,
  mismatch_type
FROM classified
WHERE mismatch_type IS NOT NULL
ORDER BY excess_aws_usd DESC, observation_id;
```